

CONNECTION FORMS

NISHTHA TIKALAL

ABSTRACT. This section underscores the dual utility of differential 1-forms in framing the structural behavior of moving coordinate apparatuses. Having established that connection 1-forms offer the most elegant and minimal description of a frame field's geometric rate of rotation, the framework expands to show that the frame field itself can be intrinsically characterized using 1-forms. This perspective unifies both the position and rotational dynamics of the moving basis within a single, cohesive differential form language.

CONTENTS

1 The Structural Equations

1

1. THE STRUCTURAL EQUATIONS

Definition 1.1 (Dual 1-Forms). If E_1, E_2, E_3 is a frame field on $\mathbf{E}^3$, then the *dual 1-forms* $\theta_1, \theta_2, \theta_3$ of the frame field are the 1-forms such that

$$\theta_i(\mathbf{v}) = \mathbf{v} \cdot E_i(\mathbf{p})$$

for each tangent vector $\mathbf{v}$ to $\mathbf{E}^3$ at $\mathbf{p}$.

Because each θ_i operates linearly on the tangent vectors at any given point $\mathbf{p}$, it functions strictly as a differential 1-form. Pointwise, the set $\{\theta_1, \theta_2, \theta_3\}$ serves as the formal dual basis of the vector space spanned by $\{E_1, E_2, E_3\}$.

In the specific case of the standard, natural Euclidean frame field U_1, U_2, U_3 , the corresponding dual forms are simply the standard coordinate differentials dx_1, dx_2, dx_3 . This identity stems directly from the action of coordinate differentials on a tangent vector $\mathbf{v}$:

$$dx_i(\mathbf{v}) = v_i = \mathbf{v} \cdot U_i(\mathbf{p}) \implies dx_i = \theta_i$$

Utilizing this dual framework allows us to rewrite the fundamental orthonormal expansion formula for any general vector field V on $\mathbf{E}^3$ in an elegant, coordinate-free formulation:

$$V = \sum_{i=1}^3 \theta_i(V) E_i$$

This expression showcases the classic dual relationship where the 1-forms directly extract the scalar component fields of the vector field along each basis direction.

Lemma 1.2. *Let $\theta_1, \theta_2, \theta_3$ be the dual 1-forms of a frame field E_1, E_2, E_3 . Then any 1-form ϕ on $\mathbf{E}^3$ has a unique expression*

$$\phi = \sum_i \phi(E_i) \theta_i$$

The relationship between an arbitrary frame field and the natural background coordinate frame mirrors perfectly when transitioning to their dual spaces.

Recall that a frame field E_1, E_2, E_3 is related to the natural Euclidean frame U_1, U_2, U_3 via the component functions a_{ij} of its attitude matrix:

$$E_i = \sum_{j=1}^3 a_{ij} U_j \quad (1 \leq i \leq 3)$$

The dual 1-forms θ_i transform along the standard coordinate differentials dx_j using the exact same coefficient entries:

$$\theta_i = \sum_{j=1}^3 a_{ij} dx_j$$

Proof. By applying the unique expansion lemma for 1-forms on the natural frame field, we have:

$$\theta_i = \sum_{j=1}^3 \theta_i(U_j) dx_j$$

Using the definition of the dual 1-forms ($\theta_i(\mathbf{v}) = \mathbf{v} \cdot E_i$), we expand the inner evaluation:

$$\begin{aligned} \theta_i(U_j) &= E_i \cdot U_j \\ &= \left(\sum_{k=1}^3 a_{ik} U_k \right) \cdot U_j \\ &= a_{ij} \end{aligned}$$

This confirms that the transformation coefficients for both the vector fields and their dual 1-forms are identical. $\square$

Now that every arbitrary frame field is completely characterized by its geometric invariants—namely, its dual 1-forms θ_i and **connection 1-forms** ω_{ij} —we turn to computing their exterior derivatives $d\theta_i$ and $d\omega_{ij}$. This systematic differentiation yields two elegant sets of identities known as **Cartan's Structural Equations**.

Theorem 1.3. *Let E_1, E_2, E_3 be a frame field on $\mathbf{E}^3$ with dual forms $\theta_1, \theta_2, \theta_3$ and connection forms ω_{ij} ($1 \leq i, j \leq 3$). The exterior derivatives of these forms satisfy:*

1. The First Structural Equations:

$$d\theta_i = \sum_j \omega_{ij} \wedge \theta_j \quad (1 \leq i \leq 3)$$

2. The Second Structural Equations:

$$d\omega_{ij} = \sum_k \omega_{ik} \wedge \omega_{kj} \quad (1 \leq i, j \leq 3)$$

Example 1.4. Verify Cartan's structural equations for the spherical frame field (Example 6.2), whose dual 1-forms and connection forms are given by:

$$\begin{aligned}\theta_1 &= dp & \omega_{12} &= \cos \varphi \, d\vartheta \\ \theta_2 &= \rho \cos \varphi \, d\vartheta & \omega_{13} &= d\varphi \\ \theta_3 &= \rho \, d\varphi & \omega_{23} &= \sin \varphi \, d\vartheta\end{aligned}$$

1. *Verification of the First Structural Equation for $i = 3$.* The first structural equation states that $d\theta_3 = \sum_{j=1}^3 \omega_{3j} \wedge \theta_j$. Expanding this sum and using the skew-symmetry property $\omega_{ij} = -\omega_{ji}$ (which implies $\omega_{33} = 0$), we obtain:

$$d\theta_3 = \omega_{31} \wedge \theta_1 + \omega_{32} \wedge \theta_2 = -\omega_{13} \wedge \theta_1 - \omega_{23} \wedge \theta_2$$

Substituting the expressions for the forms yields the individual wedge products:

$$\begin{aligned}\omega_{31} \wedge \theta_1 &= -d\varphi \wedge dp = dp \wedge d\varphi \\ \omega_{32} \wedge \theta_2 &= (-\sin \varphi \, d\vartheta) \wedge (\rho \cos \varphi \, d\vartheta) = 0 \quad (\text{since } d\vartheta \wedge d\vartheta = 0)\end{aligned}$$

Summing these terms up perfectly matches the direct exterior derivative of θ_3 :

$$d\theta_3 = dp \wedge d\varphi = d(\rho \, d\varphi)$$

2. *Verification of the Second Structural Equation for $i = 1, j = 2$.* The second structural equation states that $d\omega_{12} = \sum_{k=1}^3 \omega_{1k} \wedge \omega_{k2}$. Since $\omega_{11} = \omega_{22} = 0$, this sum collapses to a single wedge product term:

$$d\omega_{12} = \omega_{13} \wedge \omega_{32} = -\omega_{13} \wedge \omega_{23}$$

Evaluating this wedge product gives:

$$\omega_{13} \wedge \omega_{23} = d\varphi \wedge (\sin \varphi \, d\vartheta) = \sin \varphi \, d\varphi \wedge d\vartheta$$

This perfectly matches the direct exterior derivative of ω_{12} :

$$d\omega_{12} = d(\cos \varphi \, d\vartheta) = d(\cos \varphi) \wedge d\vartheta = -\sin \varphi \, d\varphi \wedge d\vartheta$$

REFERENCES

- [1] B. O'Neill, *Elementary Differential Geometry*, Revised 2nd ed. Boston, MA: Academic Press, 2006.